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Cyanide phytoremediation by water hyacinths (*Eichhornia crassipes*)

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Abstract

Although cyanide is highly toxic, it is economically attractive for extracting gold from ore bodies containing only a few grams per 1000 kg. Most of the cyanide used in industrial mining is handled without observable devastating consequences, but in informal, small-scale mining, the use is poorly regulated and the waste treatment is insufficient. Cyanide in the effluents from the latter mines could possibly be removed by the water hyacinth *Eichhornia crassipes* because of its high biomass production, wide distribution, and tolerance to cyanide (CN) and metals. We determined the sodium cyanide phytotoxicity and removal capacity of *E. crassipes*. Toxicity to 5–50 mg CN L^{-1} was quantified by measuring the mean relative transpiration over 96 h. At 5 mg CN L^{-1}, only a slight reduction in transpiration but no morphological changes were observed. The EC_{50} value was calculated by probit analysis to be 13 mg CN L^{-1}. Spectrophotometric analysis indicated that cyanide at 5.8 and 10 mg L^{-1} was completely eliminated after 23–32 h. Metabolism of $K^{14}CN$ was measured in batch systems with leaf and root cuttings. Leaf cuttings removed about 40% of the radioactivity from solution after 28 h and 10% was converted to $^{14}CO_2$; root cuttings converted 25% into $^{14}CO_2$ after 48 h but only absorbed 12% in their tissues. The calculated K_m of the leaf cuttings was 12 mg CN L^{-1}, and the V_{max} was 35 mg CN (kg fresh weight)$^{-1}$ h^{-1}. Our results indicate that *E. crassipes* could be useful in treating cyanide effluents from small-scale gold mines.

Keywords: Phytoremediation; Water hyacinth (*Eichhornia crassipes*); $K^{14}CN$; Cyanide; Gold mining

1. Introduction

For more than 100 years, cyanide (CN) has been used to extract mined gold following the cyanide leaching gold recovery (CLGR) process. About 80 million kg CN is consumed in North American gold mining annually (Eisler and Wiemeyer, 2004). A highly concentrated cyanide solution is used to extract fine-grinded ore. Cyanide forms very stable complexes with gold:

$$2Au_{(s)} + 4NaCN_{(aq)} + H_2O + 1/2O_{2(aq)} \leftrightarrows 2Na[Au(CN)_2]_{(aq)} + 2NaOH_{(aq)}$$

CLGR makes it economically worthwhile to extract even very low-grade ore, i.e., containing 0.5–13.7 g gold per 1000 kg rock (Korte et al., 2000).

The industrially generated effluents from the extraction are often stored in open tailing ponds, sometimes 150 ha or larger (Eisler and Wiemeyer, 2004). This waste contains up to 120 mg L^{-1} free CN and 400 mg L^{-1} total CN, including various cyanide complexes with heavy metals (Young, 1993), e.g., copper at about 50 mg L^{-1} (Gos and Ladwig, 1992). Such a waste management practice is a latent risk for the environment (Korte et al., 2000). For example, in January 2000, the dam of a gold leaching retention basin burst in north-western Romania, near the city of Baia Mare. About 100000 kg of cyanide was released into the river environment (UNEP/OCHA, 2000).

Another environmental hazard, especially in South America, is caused by informal, small-scale mining, which comprises all non-industrial mining activities that are not under public authority control. The generated wastewaters take the "least expensive waste disposal route" and are discharged without treatment into rivers (Tarras-Wahlberg et al., 2001; Eisler and Wiemeyer, 2004).

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Cyanide occurs naturally in plant cells as a by-product in the last step of ethylene synthesis and is rapidly detoxified by reacting with cysteine to form asparagine (Manning, 1988). The enzyme β-Cyanoalanine synthase (CAS) catalyzes the conversion of cyanide and cysteine to β-cyanoalanine and sulfide. CAS is widely distributed in higher plants and plays an important role in cyanide metabolism (Miller and Conn, 1980; Hatzfeld et al., 2000; Maruyama et al., 2001). Free cyanide in microorganisms is oxidatively detoxified. Cyanide monooxygenase converts free cyanide to cyanate, which is further mineralised by cyanase to form carbon dioxide and ammonia. Cyanide dioxygenase directly mineralises cyanide (Kunz et al., 2001).

Various plants have been shown to tolerate and eliminate cyanides from nutrient solutions (Trapp et al., 2003; Larsen et al., 2004; Samiotakis and Ebbs, 2004; Yu et al., 2004; Yu et al., 2005a). However, floating aquatic plant species, which are suitable for use in constructed wetlands or wastewater treatment ponds, have not yet been tested.

The water hyacinth *Eichhornia crassipes* is a floating macrophyte that originated in tropical South America and is now widespread in all tropic climates. If it is introduced into foreign aquatic ecosystems, it could cause severe water management problems because of its vegetative reproduction and high growth rate (Gopal and Sharma, 1981; Giraldo and Garzón, 2002). However, its enormous biomass production rate, its high tolerance to pollution, and its heavy-metal and nutrient absorption capacities (Misbahuddin and Fariduddin, 2002; Trivedy and Pattanshetty, 2002; Williams, 2002; Singhal and Rai, 2003; Ghabbour et al., 2004; Jayaweera and Kasturiarachchi, 2004) qualify it for use in wastewater treatment ponds. These attributes of *E. crassipes* and its availability in the mining regions of South America make the water hyacinth an appropriate candidate for the treatment of cyanide effluents. Here we tested the CN tolerance and the CN-elimination capacity of *E. crassipes*.

2. Materials and methods

2.1. Reagents

2.1.1. Cyanide solutions

Cyanide is highly toxic and fast reacting and becomes volatile at pH values lower than 11.5. Therefore, cyanide solutions should be handled with good ventilation, and skin contact with solid or aqueous cyanide species should be avoided.

A cyanide stock solution (1.0 g L^{-1}) was prepared by dissolving 1.883 g NaCN in 1 l distilled water. The pH of the stock solution was adjusted to 11.5 with 0.1 N NaOH. HCN has a pK_a of 9.3, i.e., 99.4% is dissociated at pH 11.5. Solutions of $K^{14}CN$ were prepared by diluting a 9.25 MBq stock solution of $K^{14}CN$ with 10 mM NaOH (1.88 GBq $mmol^{-1}$; Hartmann Analytic, Braunschweig, Germany).

2.1.2. Preparation of Hoagland solution

Hoagland solution, used in all experiments, contained 0.5 mM $MgSO_4 \cdot 7H_2O$, 1.25 mM KNO_3, 1.11 mM $Ca(NO_3)_2 \cdot 4H_2O$, 0.25 mM KH_2PO_4, 11.5 μM H_3BO_3, 2.3 μM $MnCl_2 \cdot 4H_2O$, 0.19 μM $ZnSO_4 \cdot 7H_2O$, 0.09 μM $Na_2MoO_4 \cdot 2H_2O$, 0.09 μM $CuSO_4 \cdot 5H_2O$, 17.9 μM $FeCl_3 \cdot 6H_20$, and 51 μM EDTA. In experiments lacking a N source, $Ca(NO_3)_2$ and KNO_3were replaced by $CaCl_2$ and KCl. Solutions were freshly prepared with Millipore water and autoclaved. The initial pH of all cyanide–Hoagland solutions in each experiment was adjusted to pH 8.5 with NaOH.

2.2. Chemical analyses

The cyanide concentration in cyanide removal tests with entire plants was measured spectrophotometrically following the ISO 6703 procedure. Briefly, cyanide ions react with a chlorinating agent to form cyanogen chloride, which in turn reacts with 1,3-dimethybarbituric acid in the presence of pyridine to form a violet dye (Königs reaction), which is measured spectrophotometrically at 585 nm (Aldridge, 1944).

Radioactivity was quantified using a Beckman LS-5000 TD liquid scintillation counter (Beckman Coulter, Fullerton CA, USA) and the cocktail Lumasafe Plus (Lumac LSC, Groningen, The Netherlands). Quenching of radioactivity was corrected using external standards; the standard error was about 1% (Vinken et al., 2005). ^{14}CN was also determined after transformation with Ni(II)chloride to form $[Ni(^{14}CN)_4]^{2-}$, which has absorption maxima at 267, 284, and 315 nm. The complex was separated from other species by anion-exchange HPLC with gradient separation (Hewlet Packard, series 1100) on an IONPAC AS14A column. The compounds were eluted with gradients of (B) $NaClO_4$ (250 mM) in (A) H_2O at pH 11 (0%–100% B in A in 4 min, 100%–0% B in A in 2 min, followed by 6 min 100% A; flow rate: 1.0 mL min^{-1}; Karmarkar, 2002). Eluted compounds were detected with a Raytest Ramona 2000 radiodetector, and UV absorption at 267 and 230 nm was measured. This method distinguishes the radioactive and UV signals of metal–cyanide complexes with a negative charge from those of $O^{14}CN^-$ and $^{14}CO_3^{2-}$. The detection limit for free cyanide in a nickel complex was 0.1 mg L^{-1}. HPLC fractions were collected, and the radioactivity was measured again by liquid scintillation counting (Corvini et al., 2004).

2.3. Plant cultivation

Water hyacinths (*E. crassipes*) were grown vegetatively in 50 l tubs in Hoagland solution in a greenhouse under controlled environmental conditions with a 16 h light period (light intensity of 85 μmol m^{-2} s^{-1}), a 25 °C light/20 °C dark regime, and 60% relative humidity. Plants of similar age (1.5–2 months), fresh weight (approximately 35 g), root development, and number of leaves (5–7) were selected for the tests. Offshoots and damaged leaves were removed before the plants were washed with tap water and Millipore water.

2.4. Toxicity experiments and cyanide removal

Phytotoxicity was tested in an environmental chamber with 16 h light per d (light intensity of 320 μmol m^{-2} s^{-1}), a 27 °C light/16 °C dark regime, 80% relative humidity, and constant air flow. All tests and controls were replicated ($n = 5$). CN was added at 5, 10, 20, and 50 mg L^{-1}; the control lacked cyanide. All $K^{14}CN$ solutions were initially adjusted to pH 8.5 with NaOH. Toxicity was determined as a decrease in the transpiration rate (Trapp et al., 2003) by measuring the mass loss from nutrient solution at 24, 48, 72, and 96 h. The plants were placed in opaque hydroponic vessels. Sealed polyethylene caps avoided evaporation from the solution surface when the shoots were separated from the roots and the nutrient solution to ensure that mass loss was only attributed to transpiration by leaves. During the experiment, the sample vessels were lightly agitated at 70 rpm. In the first 24 h, the plants were only exposed to Hoagland solution (transpiration $t_0 = 100\%$). The nutrient solution of plants with similar absolute transpiration was replaced with 600 mL cyanide solution. The mean relative transpiration RT was calculated as follows:

$$\mathrm{RT} = \frac{1}{n}\sum_{i=1}^{n}\frac{T_i(C,t)}{T_i(C,0)} \times 100,$$

where C is the cyanide concentration (mg CN L^{-1}), t is time (h), T is the absolute transpiration (g h^{-1}), and i is replicate 1, 2, ..., n. Water hyacinths have a high water demand and transpiration rate. The transpiration rate is a very sensitive toxicity parameter because stress factors, e.g., toxic substances, have a high impact on the water balance of these plants. Therefore, chemical stress induces a rapid decrease in transpiration and a desiccation of leaves.

Cyanide removal was tested in the same experimental set-up. The influence of the light/dark regime and the use of CN as the only N source was tested with 5.1 mg CN L^{-1} with and without nitrate (the only N source) in Hoagland solution. Controls (4) to test losses of CN by microbial degradation or volatilisation had an initial concentration of 5.2 mg CN L^{-1} and lacked plants. Test and control samples (1 mL) were diluted with Millipore water to 6 mL before photometric detection. The pH was measured after termination of the assay.

2.5. ^{14}CN removal in closed batch systems

All tests were carried out in closed batch systems in triplicate. The control to quantify the effects of volatilisation, hydrolysis, microbial degradation, photolysis, and handling contained cyanide and $K^{14}CN$, but lacked plant tissue.

Leaves of the water hyacinth were washed with Millipore water and isopropyl alcohol; roots were washed with Millipore water. Leaf or root cuttings (0.5 g) were placed in 100 mL transparent (leaves and control) and opaque (roots) glass flasks containing 15 mL autoclaved Hoagland solution, 10 mg CN L^{-1}, and 0.33–0.5 MBq $K^{14}CN$. The pH was adjusted to 8.5 with 1 N NaOH. The flasks were incubated at 25 °C with an additional light source (light intensity of 85 μmol m^{-2} s^{-1}). Evolved HCN, $H^{14}CN$, and $^{14}CO_2$ were fixed in traps filled with 3 mL 0.1 N NaOH (compare with Heim et al., 1994). The ^{14}CN in solution was measured at intervals, depending on the cyanide removal rate. To exclude the radioactivity of $^{14}CN^-$ from $^{14}CO_2$ in the traps, carbonate was precipitated with $BaCl_2$. The precipitate was washed three times with 0.01 N NaOH and suspended in scintillation cocktail for liquid scintillation counting. Samples of the NaOH solution in the traps were also analysed via HPLC. At the end of the experiments, leaf and root cuttings were ground under liquid nitrogen, sonicated for 1 h, and extracted for 16 h with 2.5 M NaOH in the dark (Bushey et al., 2004). The total radioactivity and cyanide concentration in the extract were measured by liquid scintillation counting and anion-exchange HPLC. The solid residue obtained after filtration was washed with 0.01 N NaOH, dried, and combusted in an oxidizer at >800 °C. The evolving $^{14}CO_2$ was absorbed in CarboMax Plus (Lumac-LSC, Groningen, Netherlands), and the radioactivity was measured by liquid scintillation counting.

2.6. Determination of Michaelis–Menten parameters

To calculate the rate of cyanide disappearance from the nutrient solution, volatilisation of HCN must be avoided. To achieve this, a second series of batch experiments were performed in closed tubes with no head space above the solution. Leaf cuttings (0.5 g) were placed in 20 mL tubes with 20 mL Hoagland solution and sealed with a gas-tight membrane. The pH was adjusted to 8.5. The tubes were kept at a constant temperature of 25 °C. Cyanide was added to 2, 4, 6, 10, and 16 mg L^{-1}. Controls to account for the losses via volatilisation, hydrolysis, microbial degradation, photolysis, and handling contained 2 and 16 mg CN L^{-1}, but lacked leaves. All solutions contained 0.259 MBq $K^{14}CN$. Samples were taken at 0, 2, 4, 6, and 8 h through the membrane with a Hamilton syringe (three replicates for each concentration) without opening the tubes. The pH was measured after termination of the experiment. The Michaelis–Menten parameters were finally calculated with the 4 h data point, where the best fittings were achieved.

2.7. Statistical analysis

Each measurement was performed in triplicate ($n = 3$), except for the toxicity experiments, which were performed in five parallels ($n = 5$). The difference between specific pairs of means was identified by a two-sided T-test (significance level: $P < 0.05$). EC_x values, resulting of the toxicity tests, including F-tests for regression (significance level: $P < 0.05$), were calculated with the ToxRat Professional Version 2.09 (ToxRat Solutions). Michaelis–Menten parameters, including regression analysis of variance (significance level: $P < 0.05$), were calculated by non-linear regression and from a Lineweaver–Burk plot using SigmaPlot (Windows Version 8.0).

3. Results

3.1. Toxicity experiments and cyanide removal

In the cyanide toxicity experiments, cyanide decreased the relative transpiration of *E. crassipes*; higher concentrations led to lower transpiration (Fig. 1). At the lowest concentration (5 mg L^{-1}), transpiration was only slightly reduced and the plants survived without any morphological changes. At higher concentrations, the plants developed chlorosis and the leaves faded after 48 h. All plants at 50 mg CN L^{-1} and 60% of the plants at 20 mg CN L^{-1} died during the experiment; plants exposed to 10 mg CN L^{-1} survived, but lost about 50% of their leaves by desiccation. The relative transpiration of the control plants did not change from the initial relative transpiration, illustrating the fitness of *E. crassipes* in these unusual surroundings.

The effective concentrations (EC) were calculated using the relative transpiration at the end of the experiment

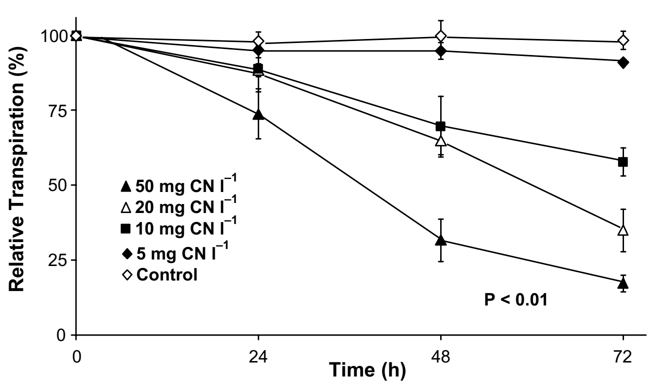

Fig. 1. Relative transpiration of *E. crassipes* growing in hydroponic solution supplied with cyanide at different concentrations. Transpiration is expressed relative to the initial transpiration of each plant (=100%) without cyanide treatment. Error bars denote standard deviation ($n = 5$). P-values are derived from T-tests. Results were compared with the relative transpiration at the end of the experiment (72 h).

(72 h) by linear regression with probit and logit analyses. Both methods showed significant fits ($P < 0.04$) (Table 1).

In the cyanide removal experiments with entire plants, the plants removed significantly higher amounts of cyanide than the control ($P \leqslant 0.01$) (Fig. 2). The cyanide concentration in the controls did change from the initial concentration. The pH in all plant treatments decreased to 5.9–6.6, whereas the pH in the controls remained about 8.5. Using the pK_a (HCN) of 9.3, we calculated a decrease in the content of free cyanide ions from 13.4% to <0.2%, due to the formation of HCN.

In the first experimental dark/light period 1 with a short initial light period at 27 °C (Fig. 2, filled symbols; p1: 3 h light, 8 h dark, 16 h light), the concentration of cyanide at initial concentrations of 5.8 and 10 mg L^{-1} did not change much during the first 12.5 h; at this time, only about 1 mg CN L^{-1} was removed. However, the remaining cyanide in solution was rapidly removed during the next 12 h (5.8 mg CN L^{-1}) or 16 h (10 mg CN L^{-1}).

To determine whether this lag phase in cyanide removal is affected by the dark period at a 11 °C lower temperature, we carried out another experiment with longer light period at 16 °C with cyanide at a concentration of 5.1 mg L^{-1} (Fig. 2, open symbols; p2: 11.5 h light, 8 h dark, 16 h light). Only approximately 1.6 CN L^{-1} was removed during the first 23 h, i.e., the lag phase was at least 10 h longer than the lag phase in the experiment with a short initial light period. A further 1.6 CN L^{-1} was rapidly removed during the next 5 h. We also examined the influence of cyanide as the only N source (Fig. 2, open squares), but no differences were observed ($P = 0.32$). After extended incubation (19 h), the cyanide in both treatments was completely removed (data not shown).

3.2. ^{14}CN removal in closed batch systems

We determined cyanide removal by leaf cuttings incubated under light (Fig. 3, open symbols) and in the dark (filled symbols). Similar first-order removal kinetics were observed under the two conditions, and ^{14}CN was removed to 10% of the initial concentration after 24 h. The slightly lower ^{14}CN concentration in the sample under light compared to the sample in the dark (Fig. 3) was caused by an additional sampling and hence a lower remaining volume of the sample under light. The ^{14}CN content in the control lacking plant cuttings decreased to 27% after 48 h (Fig. 3). We attributed this cyanide loss to sampling and volatilisation of $H^{14}CN$. The ^{14}CN recovery in the control

Table 1
Results of toxicity tests – calculated EC_x values (mg CN L^{-1}) for *E. crassipes* in hydroponic solution

Method	EC_{50} (mg CN L^{-1})	EC_{20} (mg CN L^{-1})	EC_{10} (mg CN L^{-1})	P	R^2
Probit linear regression	13.0	4.7	2.7	0.034	93.2
Two-parameter logistic linear regression	12.8	4.6	2.6	0.037	92.8

Values were determined using the relative transpiration of *E. crassipes* at the end of the experiment after 72 h (see Fig. 1). P-values are derived from F-test for regression.

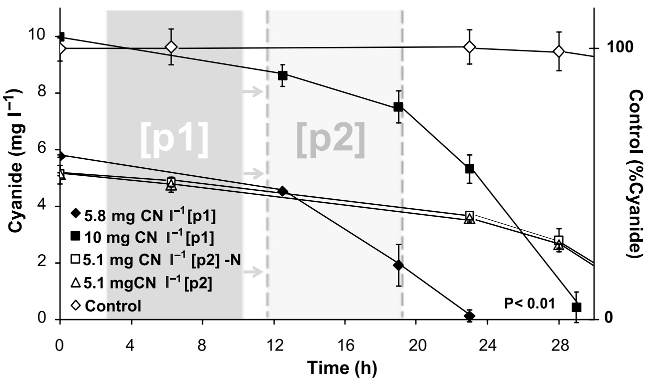

Fig. 2. Cyanide removal by entire water hyacinth plants in hydroponic vessels. p1 (filled symbols): dark/light period 1 (27 °C; 3 h light, 8 h dark, 16 h light); p2 (open symbols): dark/light period 2 (16 °C; 11.5 h light, 8 h dark, 16 h light); –N: cyanide as only N source (right y-axis: the cyanide concentration of the control lacking plants is expressed in % relative to the initial CN concentration). The P-value is derived from a T-test; results of the different treatments were compared with the control. Error bars denote standard deviation ($n = 3$).

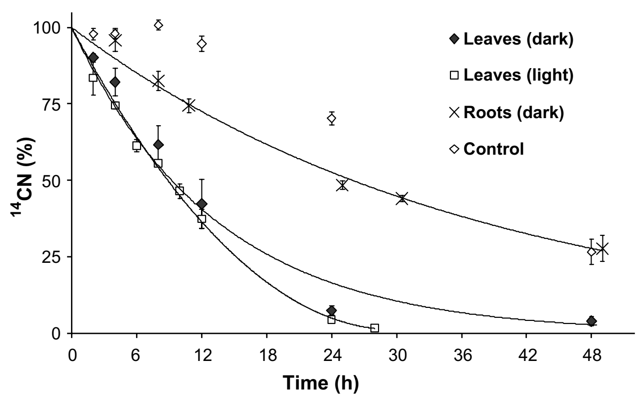

Fig. 3. ^{14}CN removal in closed batch systems with root and leaf cuttings of *E. crassipes*. Volatilised $^{14}CO_2$ and $H^{14}CN$ were fixed in 0.1 N NaOH traps. The P-value is derived from non-linear regression of the exponentially fitted decay curves. Error bars denote standard deviation ($n = 3$).

was about 100%, and only 0.15% was converted to $^{14}CO_2$ (Table 2).

Almost 35%–40% of the applied radioactivity in both conditions was found in the leaf extract and 6%–10% was converted to $^{14}CO_2$ (Table 2). No UV or radioactive signals of free ^{14}CN and related cyanide species were detected in the extracts. Calculated with the detection limit of 0.1 mg CN L^{-1} the total content of ^{14}CN in the leaf tissue extract must below 3.3%. Only a small amount was non-extractable and found in the residue fraction (Table 2).

Root cuttings were incubated only in the dark, i.e., under natural conditions. Cyanide removal by root cuttings was lower than that by leaves (Fig. 3). After 24 h, 48% of the ^{14}CN remained in solution, and at the end of the experiment (49 h), 28% remained. About 12% of the applied radioactivity was found in the root extracts, and no signals of free or metal-complexed ^{14}CN were detected. Approximately 25% of the applied ^{14}CN was mineralised to $^{14}CO_2$ (Table 2).

3.3. Determination of Michaelis–Menten parameters

In our experiments, volatilisation of ($H^{14}CN$) was avoided. The ^{14}CN recovery in the controls was 100.1 ± 4.3% (2 mg CN L^{-1}) and 99.1 ± 2.1% (16 mg CN L^{-1}). The initial pH of the solution was adjusted to 8.5. At the end of the experiment, the pH decreased to 6.8–7.2. Using the pK_a of HCN of 9.3, we calculated that the fraction of the cyanide ion (CN^-) in solution decreased from 13.4% to 0.8–0.3% during the experiment.

The Michaelis–Menten parameters V_{max} (maximum velocity of cyanide removal) and K_m were determined by linear regression with a Lineweaver–Burk plot and by non-linear regression. The results obtained were similar [V_{max}: 35.5 ± 4.3 and 35.0 ± 2.4 mg kg (fresh weight)$^{-1}$ h^{-1}; K_m: 12.0 ± 1.8 and 11.8 ± 1.5 mg L^{-1}, respectively] and significant ($P < 0.01$; Fig. 4).

4. Discussion

In our experiments, the water hyacinth *E. crassipes* was more tolerant towards free cyanide in solution than other plants tested by others. Cyanide at 2 mg L^{-1} caused a 50% decrease in normalised relative transpiration of the basket willow *Salix viminalis* after 72 h, and 8 mg L^{-1} killed the plants, yet the cyanide concentration in all treatments was reduced to approximately 0% after 190 h (Larsen et al., 2004). Similar results were obtained with the weeping willow *Salix babylonica* (Yu et al., 2005a).

Table 2
^{14}C-balance of ^{14}CN removal in closed batch systems with leaf and root cuttings of *E. crassipes*

Plant compartment	Plant extract (%)[a]	$^{14}CO_2$ trapped (%)[a]	$H^{14}CN$ trapped (%)[a]	Non-extractable ^{14}C-residue (%)[a]	Remaining in solution (%)[a]	Sampling (%)[a]	Sum (%)[a]
Leaves (light)	35.5 ± 2.2	10.5 ± 1.7	15.4 ± 5.3	3.8 ± 0.6	1.5 ± 0.3	24.8 ± 3.4	91.5 ± 1.3
Leaves (dark)	37.0 ± 6.5	6.4 ± 1.2	25.1 ± 2.4	1.3 ± 0.2	3.9 ± 1.6	16.0 ± 0.9	89.7 ± 4.8
Roots	11.9 ± 0.8	24.9 ± 2.4	26.9 ± 5.5	1.1 ± 0.5	27.6 ± 4.2	2.5 ± 0.1	97.9 ± 1.8
Control	–	0.1 ± 0.03	59.9 ± 2.6	–	26.6 ± 4.1	20.7 ± 0.4	107.4 ± 4.8

Gaseous ^{14}C species ($^{14}CO_2$,$H^{14}CN$) were fixed in 0.1 N NaOH traps. Plants were homogenised by grinding with liquid nitrogen and sonication and then extracted with 2.5 N NaOH. Non-extractable residue was completely combusted to $^{14}CO_2$, trapped, and measured by liquid scintillation. Errors denote standard deviation ($n = 3$).

[a] Values are expressed as the % of the initial applied radioactivity.

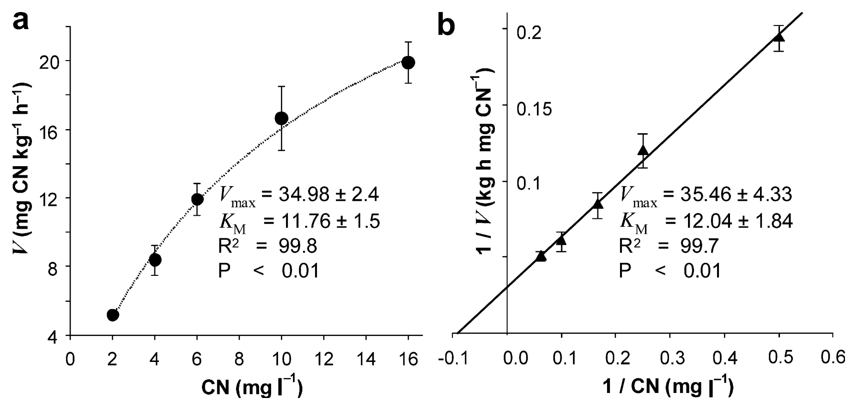

Fig. 4. Determination of Michaelis–Menten parameters of ^{14}CN removal by leaf cuttings of water hyacinths. (a) Non-linear regression, (b) Lineweaver–Burk plot. Closed tubes contained cyanide (five concentrations, $n = 3$) in with half-concentrated Hoagland solution, adjusted to pH 8.5 at 21.5 °C. Volatilisation of $H^{14}CN$ was prevented. The ^{14}CN recovery in the controls was $100.1 \pm 4.3\%$ (2 mg CN L^{-1}) and $99.1 \pm 2.1\%$ (16 mg CN L^{-1}). *P*-values were derived from regression analysis of variance. Error bars denote standard deviation ($n = 3$).

Cyanide below 1 mg L^{-1} had no strong impact on the normalised relative transpiration. At 3.72 mg CN L^{-1}, the normalised relative transpiration decreased, but the absolute transpiration slightly increased. Cyanide up to a concentration of 3.72 mg CN L^{-1} was completely eliminated; 90% of the cyanide at 9.3 mg L^{-1} was removed after 192 h in solution. The calculated EC_{50} values of cyanide for *S. babylonica* are 8.2 mg CN L^{-1} (non-linear regression) and 11.5 mg CN L^{-1} (linear regression), and thus close to our findings with *E. crassipes*. The cyanide removal capacity of *E. crassipes* and the two willow species were similar, but *E. crassipes*, unlike the willows, was able to remove cyanide in 10 mg L^{-1} solutions completely.

In our experiments, the measured pH decrease indicated that HCN was the dominant cyanide species in solution. HCN is membrane permeable without active transport processes. Ions are transported via proteins into plant cells; whether protein-mediated transport influences the uptake of CN into plant cells depends on the concentration of cyanide ions with negative charge. Felle and Hanstein (2002) injected 260 and 26 mg CN L^{-1} solution directly through the stomata of the faba bean *Vicia faba* and measured the apoplastic pH. Only the more concentrated CN solution caused a pH shift in the apoplast from pH 5.2–6.2. These results and the pH values we measured for *E. crassipes* indicate that CN at concentrations of 1–16 mg L^{-1} most likely would be present in the form of HCN in the apoplast. Therefore, protein-mediated transport of CN^- through the membrane of *E. crassipes* should be negligible.

We think that the observed prolonged lag phase in cyanide degradation with entire plants was caused by the temperature drop in the dark period and not by the duration of the light period because we did not find a light dependency in the tests with leaf cuttings. This hypothesis is supported by the results of Yu et al. (2005b), who determined a substantial temperature dependency of cyanide removal by woody plants. Another possible explanation for the prolonged lag phase is the light-dependent volatilisation of HCN through the leaves. *Salix alba* removes more cyanide from nutrient solution when exposed to light, and the concentration of HCN in the gas phase increases (Trapp et al., 2001). However, this does not mean that a light-dependent volatilisation of HCN by *E. crassipes* with transpiration as the main detoxification occurs. Furthermore, volatilisation of HCN does not provide a valid explanation for the lag phase in the experiment with an initial light period of 11.5 h. This phenomenon may point to a physiological adaptation period for CN removal in the plants and/or the CN translocation from the roots to the leaves. Time-dependent experiments with radio-labelled cyanide in flow-through systems are required to test these hypotheses.

The removal of cyanide from solution by leaf cuttings of *E. crassipes* was similar to the findings of others. The high ^{14}CN recovery of the controls indicated that the determined V_{max} and K_m were attributed to absorption and metabolic processes by plant tissue. Larsen et al. (2004) found the best cyanide removal in tests with *Salix viminalis* and *Salix schwerinii*. Leaf cuttings reduced the cyanide concentration in 18 h to 10% of the initial concentration. The calculated V_{max} and K_m were about 10 mg CN kg^{-1} h^{-1} and 1 mg CN L^{-1}, respectively. The same group found similar results with *S. viminalis* (Larsen et al., 2005). The best results were obtained with leaf cuttings; 2 g of plant material needed about 20 h to degrade 50% of the cyanide applied and >50 h to eliminate the cyanide completely at 2 mg L^{-1}. They calculated a V_{max} of 6.9 mg CN kg^{-1} h^{-1} and a K_m of 0.6 mg CN L^{-1}. One has to keep the theoretical nature of V_{max} in mind; we clearly observed that cyanide concentrations above 10 mg L^{-1} killed entire plants. This impact of toxicity was also observed in determinations of the Michaelis–Menten parameters starting with 20 mg CN L^{-1} (data not shown). As a consequence, we calculated the parameters with a maximum cyanide concentration of 16 mg L^{-1}, at which CN removal was not affected. We found no evidence for microbial or chemical degradation of cyanide in the controls lacking plant tissue.

Since we found radioactivity in the plant extracts and since no cyanide or related metal complexes were detected, cyanide must be metabolised in the plant cells. One possibility is that cyanide is converted to asparagine (Manning, 1988; Hatzfeld et al., 2000; Maruyama et al., 2001). Because of the notable production of $^{14}CO_2$ and the negligible cyanide removal in the controls, we suppose that the CN metabolite asparagine would then be mineralised in the plant cells to CO_2. Although we washed the leaves with isopropyl alcohol before transfer to autoclaved Hoagland solution to avoid microbial degradation processes, microbial contamination can only be absolutely excluded by working with sterile cell cultures. Another possibility on the fate of cyanide is the direct oxidation of CN to CO_2 via monooxygenase or dioxygenase systems, but this is unlikely since direct oxidation by plant cells has not been reported in the literature. Another possibility is the photooxidation of cyanide to form cyanate (Nowakowska et al., 1999), which could be further oxidized to CO_2 by cyanases; these enzymes have been identified in plants (Guilloton et al., 2002).

We carried out the cyanide removal experiments with roots not washed with isopropyl alcohol since wanted to test the complete root system, including rhizospheric microorganisms. Water hyacinth roots are well known for their complex rhizospheric microflora, which positively influence the uptake of nutrients and pollutants (Sipaúba-Taveres et al., 2002; So et al., 2003). Therefore, we expected that more cyanide would be removed by the roots than by the leaves. Surprisingly, cyanide removal by roots was slower and not as effective as removal by leaves. Similar results with willows were also obtained by Larsen et al. (2005). Since $^{14}CO_2$ was produced, we suppose that the rhizospheric microorganisms have a high oxidative activity.

5. Conclusions

E. crassipes demonstrated a high tolerance to cyanide and removed free cyanide in solution in short time periods, either alone or in conjunction with associated microorganisms. Other promising attributes of the water hyacinth are metal tolerance and absorption (Misbahuddin and Fariduddin, 2002; Williams, 2002; Ghabbour et al., 2004), high biomass production with good root development, low maintenance, and ready availability in mining regions. Water hyacinths are therefore suitable for use in wastewater treatment ponds. However, further experiments with other cyanide species present in mining wastewaters must be performed to test the suitability of *E. crassipes* in treating cyanide effluents from gold mining. However, caution should be practiced and the water hyacinths should not be introduced into foreign tropic aquatic ecosystems, where they might cause serious water management problems. Since cyanide in aquatic ecosystems is fatal for fishes in the ppb range, the risk of using water hyacinths in closed and controlled cyanide treatment ponds in regions where the water hyacinth is already present should be acceptable.

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